

PRAKTIKUM SISTEM OPERASI

MODUL 15

Keamanan Linux



Oleh:

Dimastian Aji Wibowo

2311104058

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

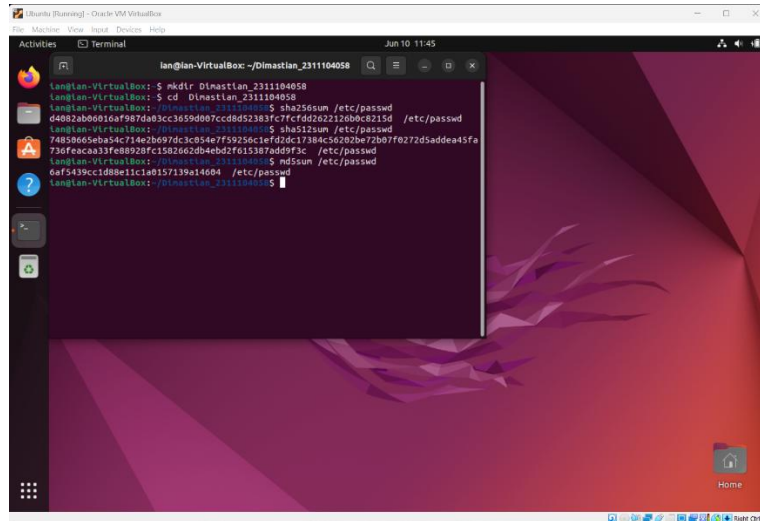
UNIVERSITAS TELKOM

2026

JURNAL

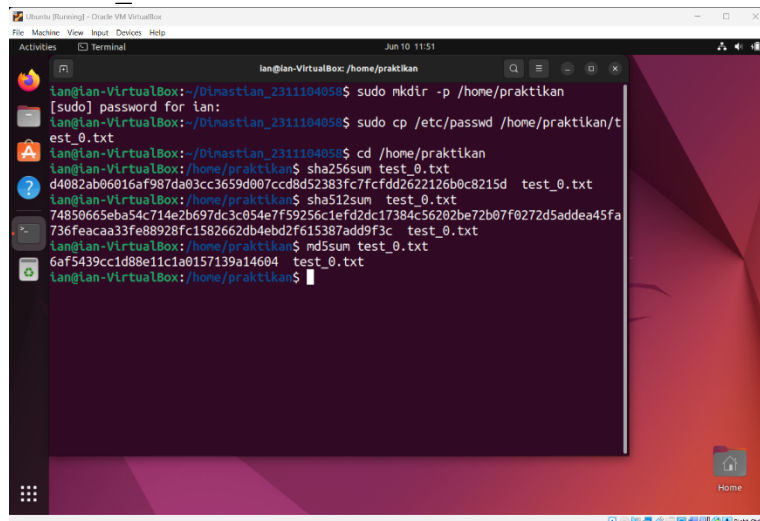
1. Integritas: Dasar Hashing

a. hash SHA256, SHA512, dan MD5



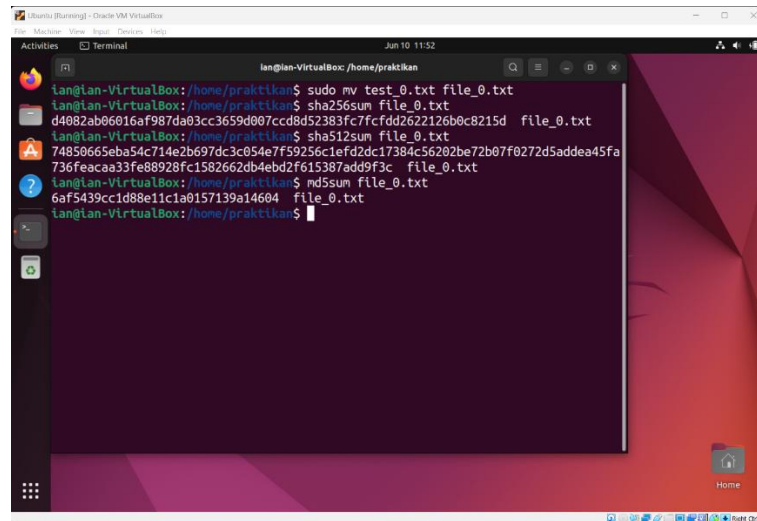
```
ian@ian-VirtualBox: ~/Dinastian_2311104058
ian@ian-VirtualBox:~$ mkdir Dinastian_2311104058
ian@ian-VirtualBox:~$ cd Dinastian_2311104058
ian@ian-VirtualBox:~/Dinastian_2311104058$ sha256sum /etc/passwd
d4082ab0616af987da03cc3659d007ccd8d52383fc77cfd2622126b0c8215d /etc/passwd
ian@ian-VirtualBox:~/Dinastian_2311104058$ sha512sum /etc/passwd
74850665eba54c714e2b697dc3c054e7f59256c1efd2dc17384c56202be72b07f0272d5addea45fa /etc/passwd
ian@ian-VirtualBox:~/Dinastian_2311104058$ md5sum /etc/passwd
6af5439cc1d88e11c1a0157139a14604 /etc/passwd
ian@ian-VirtualBox:~/Dinastian_2311104058$
```

b. File test_0.txt



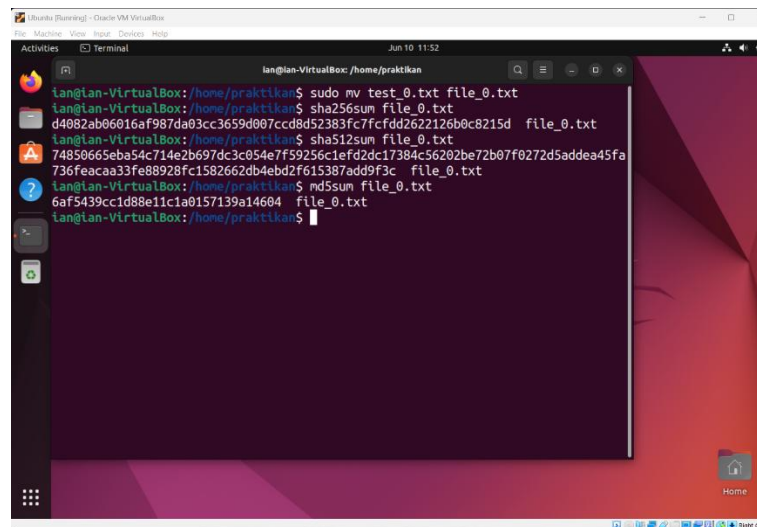
```
ian@ian-VirtualBox:~/Dinastian_2311104058$ sudo mkdir -p /home/praktikan
[sudo] password for ian:
ian@ian-VirtualBox:~/Dinastian_2311104058$ sudo cp /etc/passwd /home/praktikan/test_0.txt
ian@ian-VirtualBox:~/Dinastian_2311104058$ cd /home/praktikan
ian@ian-VirtualBox:~/home/praktikan$ sha256sum test_0.txt
d4082ab0616af987da03cc3659d007ccd8d52383fc77cfd2622126b0c8215d test_0.txt
ian@ian-VirtualBox:~/home/praktikan$ sha512sum test_0.txt
74850665eba54c714e2b697dc3c054e7f59256c1efd2dc17384c56202be72b07f0272d5addea45fa test_0.txt
ian@ian-VirtualBox:~/home/praktikan$ md5sum test_0.txt
6af5439cc1d88e11c1a0157139a14604 test_0.txt
ian@ian-VirtualBox:~/home/praktikan$
```

- c. hash SHA256, SHA512, dan MD5



```
ian@ian-VirtualBox: /home/praktikan$ sudo mv test_0.txt file_0.txt
ian@ian-VirtualBox: /home/praktikan$ sha256sum file_0.txt
d4082ab06016af987da03cc3659d007ccd8d52383fc7fcfd2622126b0c8215d file_0.txt
ian@ian-VirtualBox: /home/praktikan$ sha512sum file_0.txt
74850665eba54c714e2b697dc3c054e7f59256c1efd2dc17384c56202be72b07f0272d5addea45fa
736feacaa33fe88928fc1582662db4ebd2f615387add9f3c file_0.txt
ian@ian-VirtualBox: /home/praktikan$ md5sum file_0.txt
6af5439cc1d88e11c1a0157139a14604 file_0.txt
ian@ian-VirtualBox: /home/praktikan$
```

- d. Rename file

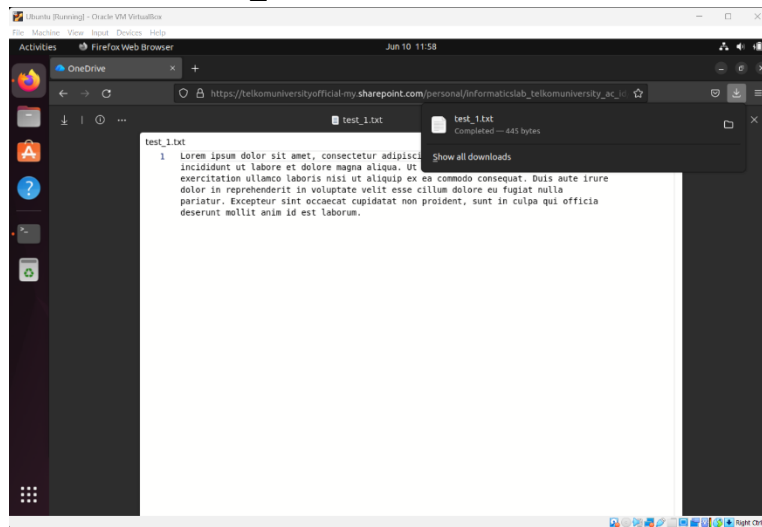


```
ian@ian-VirtualBox: /home/praktikan$ sudo mv test_0.txt file_0.txt
ian@ian-VirtualBox: /home/praktikan$ sha256sum file_0.txt
d4082ab06016af987da03cc3659d007ccd8d52383fc7fcfd2622126b0c8215d file_0.txt
ian@ian-VirtualBox: /home/praktikan$ sha512sum file_0.txt
74850665eba54c714e2b697dc3c054e7f59256c1efd2dc17384c56202be72b07f0272d5addea45fa
736feacaa33fe88928fc1582662db4ebd2f615387add9f3c file_0.txt
ian@ian-VirtualBox: /home/praktikan$ md5sum file_0.txt
6af5439cc1d88e11c1a0157139a14604 file_0.txt
ian@ian-VirtualBox: /home/praktikan$
```

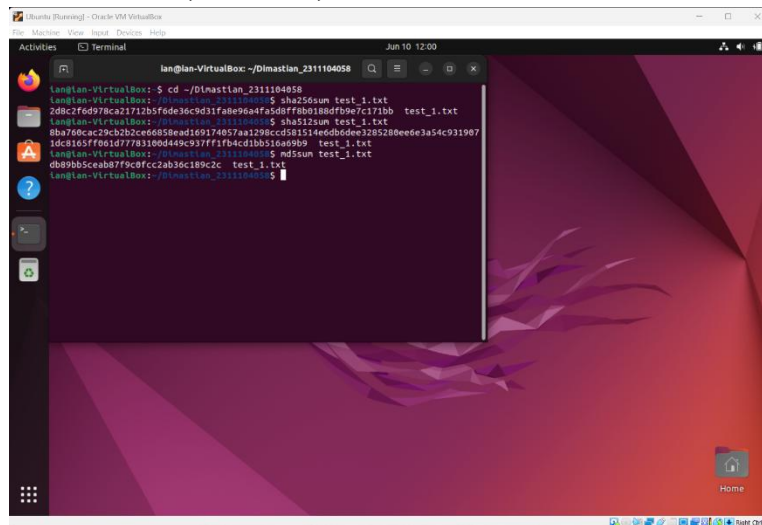
- e. File /etc/passwd, test_0.txt, dan file_0.txt akan memiliki nilai hash yang persis sama. Penjelasan, algoritma hash menghitung nilainya murni berdasarkan isi (konten/data) file, bukan berdasarkan nama atau lokasi direktorinya. Selama satu bit pun isi datanya tidak berubah, nilai integritas (hash) akan selalu identik.

2. Integritas: Avalance

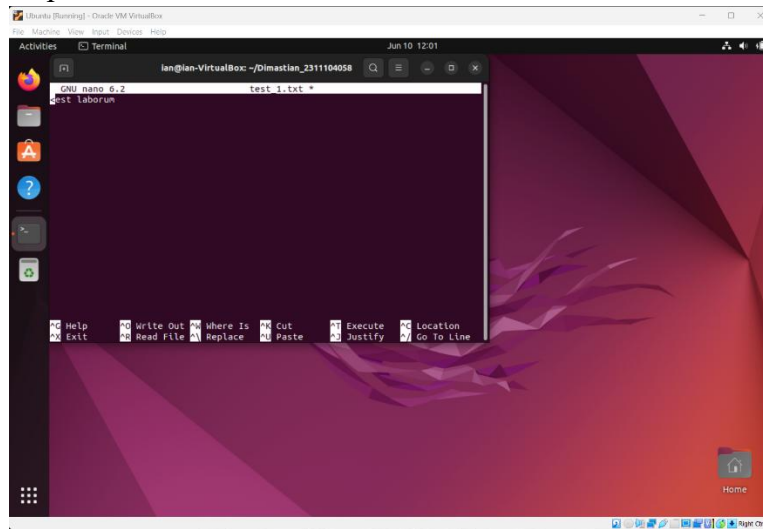
a. Download file test_1.txt



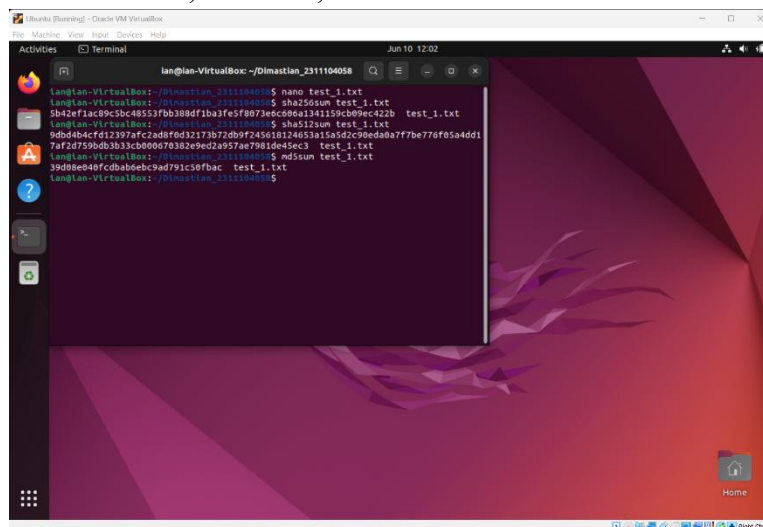
b. hash SHA256, SHA512, dan MD5



c. Hapus titik di akhir



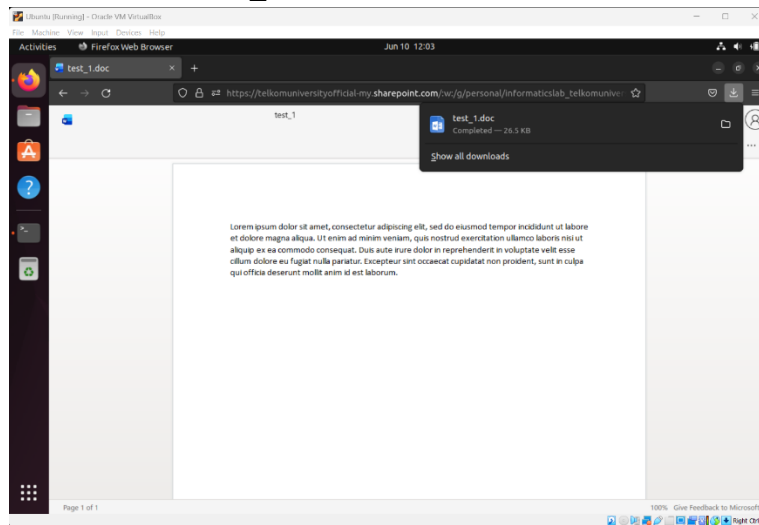
d. hash SHA256, SHA512, dan MD5



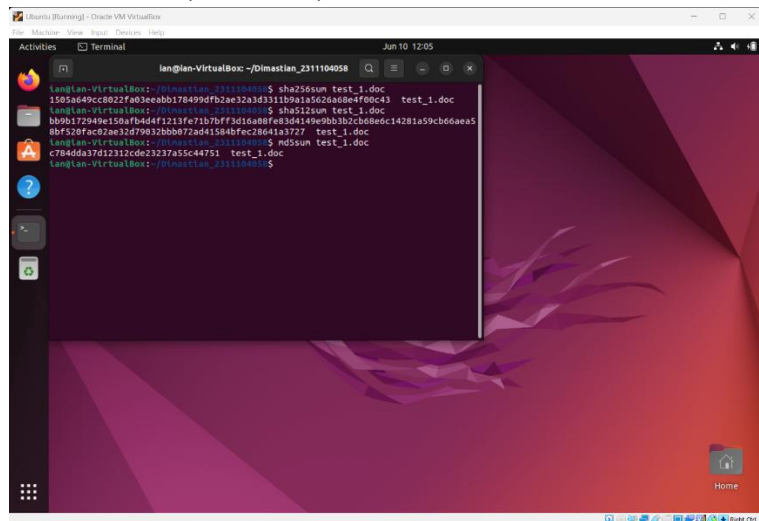
- e. Nilai hash akan berubah total atau sangat berbeda. Meskipun modifikasinya sangat kecil (hanya menghilangkan sebuah tanda titik), algoritma kriptografi bekerja melalui apa yang disebut Avalanche Effect. Setiap perubahan satu karakter/bit sekecil apa pun akan menghasilkan keluaran (hash) yang benar-benar baru. Hash yang tidak cocok menandakan bahwa integritas asal data sudah berubah/terkompromi.

3. Integritas: Metadata

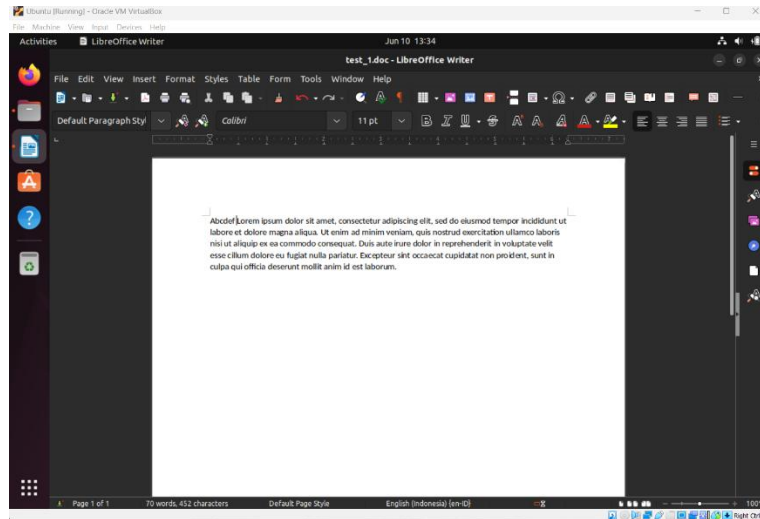
a. Download file test_1.doc



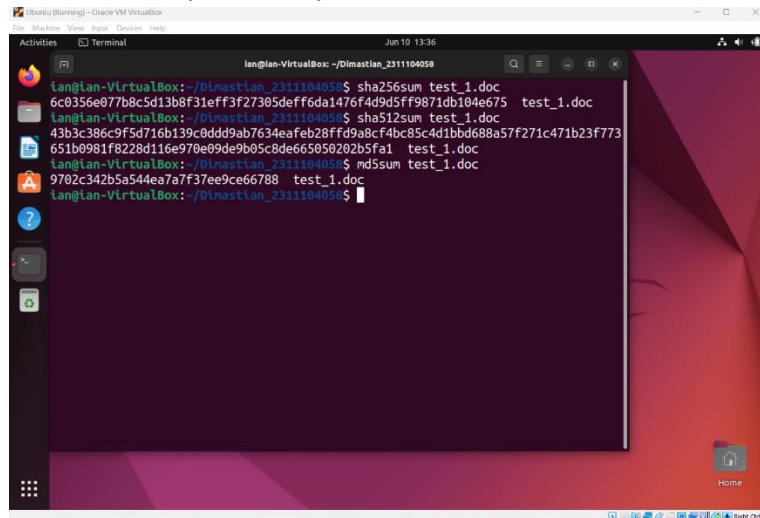
b. hash SHA256, SHA512, dan MD5



c. Ketik huruf



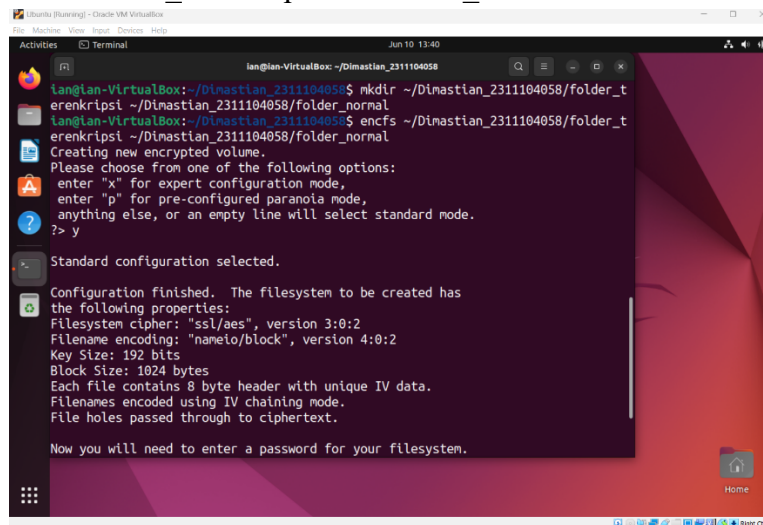
d. hash SHA256, SHA512, dan MD5



- e. Nilai hash yang baru akan menjadi berbeda dari aslinya, walaupun teks di dalamnya sekilas sudah dikembalikan seperti awal. Hal ini terjadi karena format file seperti .doc menyimpan lebih dari sekadar tulisan, ia juga menyimpan data latar belakang berupa Metadata (misal: atribut waktu file dimodifikasi, siapa penulisnya, dll). Setiap kali tombol Save ditekan, atribut "waktu modifikasi terakhir" (last modified timestamp) akan terbaru. Perubahan waktu save ini akan mengubah deretan bit dasar file yang memicu perubahan pada nilai hash yang dihasilkan.

4. **Konfidensialitas: encfs**

a. encfs folder_terenkripsi dan folder_normal



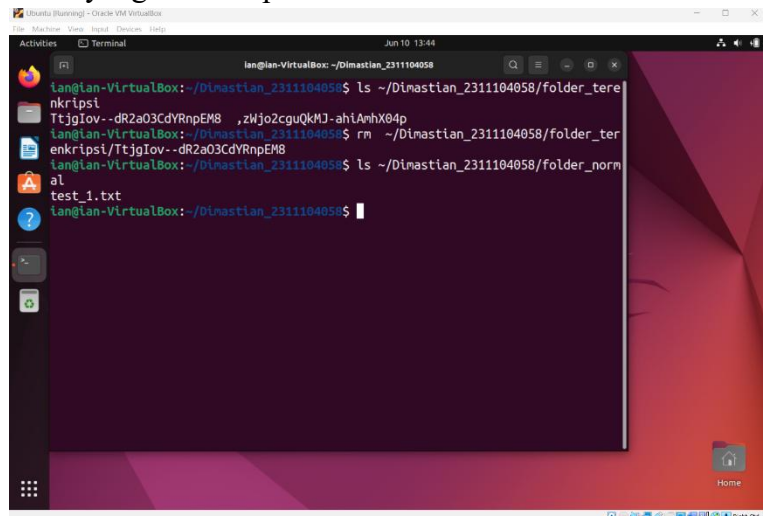
```
ian@ian-VirtualBox: ~/Dimastian_2311104058
ian@ian-VirtualBox:~/Dimastian_2311104058$ mkdir ~/Dimastian_2311104058/folder_t
erenkripsi ~/Dimastian_2311104058/folder_normal
ian@ian-VirtualBox:~/Dimastian_2311104058$ encfs ~/Dimastian_2311104058/folder_t
erenkripsi ~/Dimastian_2311104058/folder_normal
Creating new encrypted volume.
Please choose from one of the following options:
  enter "x" for expert configuration mode,
  enter "p" for pre-configured paranoia mode,
  anything else, or an empty line will select standard mode.
?> y
Standard configuration selected.
Configuration finished. The filesystem to be created has
the following properties:
Filesystem cipher: "ssl/aes", version 3:0:2
Filename encoding: "nameio/block", version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
Filenames encoded using IV chaining mode.
File holes passed through to ciphertext.

Now you will need to enter a password for your filesystem.
```

b. File asli otomatis memiliki nama yang diacak/terenkripsi, dan apabila filenya coba dibaca akan berupa sekumpulan aksara acak (ciphertext). Hal ini merupakan bentuk perlindungan konfidensialitas.

c. Hapus file

File aslinya di folder_normal juga ikut hilang/terhapus. Karena folder_normal pada hakikatnya tidak benar-benar menyimpan file, melainkan hanya bertindak sebagai jendela tempat file yang tidak terenkripsi ditampilkan, sementara wujud asli yang tersimpan secara sistem terletak di folder terenkripsinya.

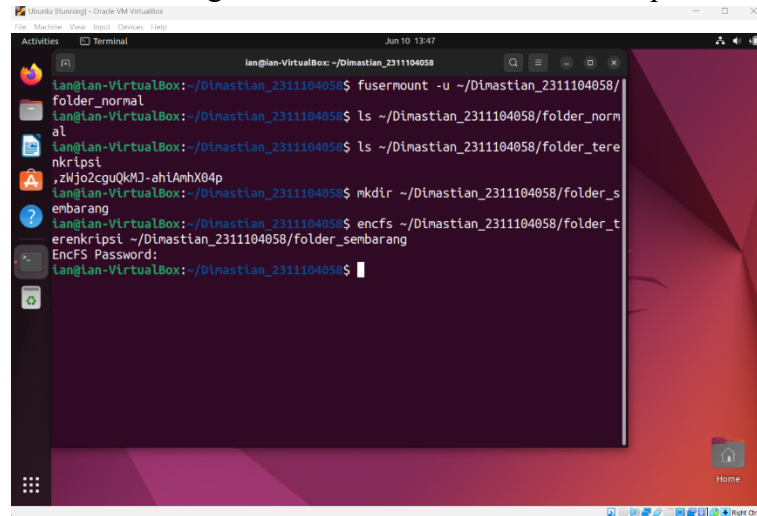


```
ian@ian-VirtualBox:~/Dimastian_2311104058$ ls ~/Dimastian_2311104058/folder_tere
nkripsi
TtjgIov--dR2a03CdYRnpEM8  ,zhJo2cguQkMJ-ahIAmhX04p
ian@ian-VirtualBox:~/Dimastian_2311104058$ rm ~/Dimastian_2311104058/folder_ter
enkripsi/TtjgIov--dR2a03CdYRnpEM8
ian@ian-VirtualBox:~/Dimastian_2311104058$ ls ~/Dimastian_2311104058/folder_norm
al
test.1.txt
ian@ian-VirtualBox:~/Dimastian_2311104058$
```

d. Unmount

folder_normal seketika berubah menjadi kosong. Seluruh file secara aman dikunci dalam bentuk tak terbaca di dalam folder_terenkripsi. File normal tersebut baru bisa dilihat dan diakses kembali setelah kamu melakukan perintah

encfs ulang dan memasukkan password yang benar

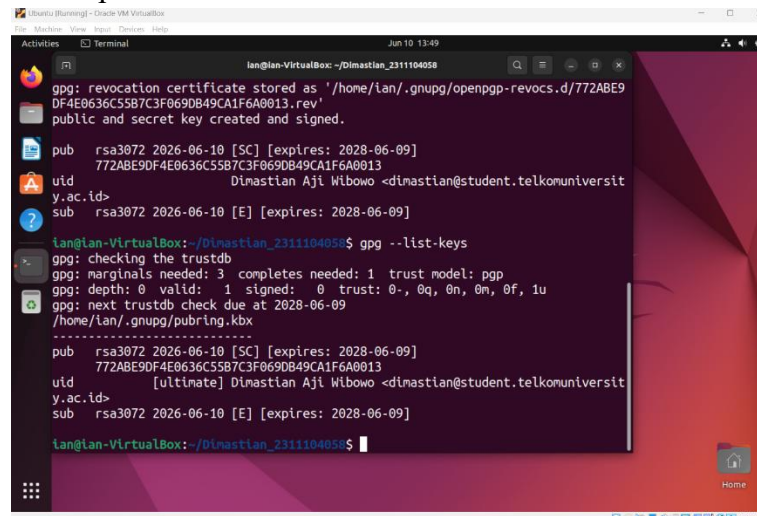


```
ian@ian-VirtualBox: ~/Dinastian_2311104058
ian@ian-VirtualBox:~/Dinastian_2311104058$ fusermount -u ~/Dinastian_2311104058/
folder_normal
ian@ian-VirtualBox:~/Dinastian_2311104058$ ls ~/Dinastian_2311104058/folder_norm
al
ian@ian-VirtualBox:~/Dinastian_2311104058$ ls ~/Dinastian_2311104058/folder_tere
nkripsi
ian@ian-VirtualBox:~/Dinastian_2311104058$ mv ~/Dinastian_2311104058/folder_tere
nkripsi ~/Dinastian_2311104058/folder_sembarang
ian@ian-VirtualBox:~/Dinastian_2311104058$ mkdir ~/Dinastian_2311104058/folder_s
embarang
ian@ian-VirtualBox:~/Dinastian_2311104058$ encfs ~/Dinastian_2311104058/folder_t
erenkripsi ~/Dinastian_2311104058/folder_sembarang
EncFS Password:
ian@ian-VirtualBox:~/Dinastian_2311104058$
```

e. folder_sembarang

5. **Konfidensialitas: gpg**

a. Kunci publik



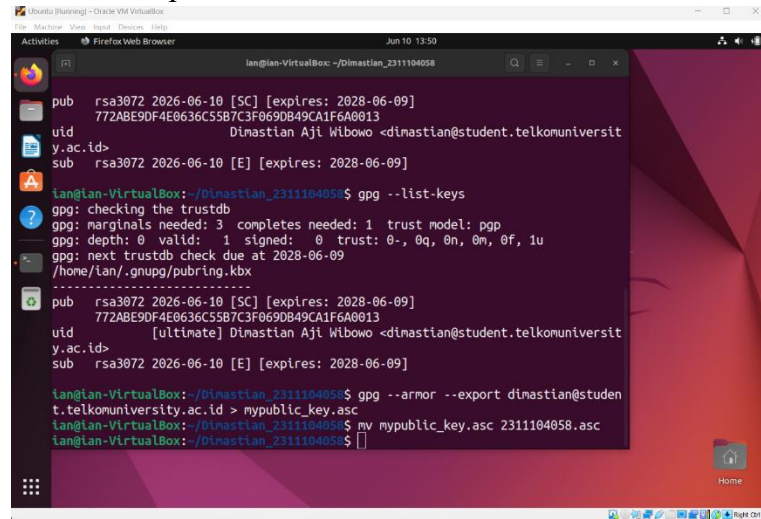
```
ian@ian-VirtualBox: ~/Dinastian_2311104058
ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --full-gen-key
gpg: revocation certificate stored as '/home/ian/.gnupg/openpgp-revocs.d/772ABE9
DF4E0636C55B7C3F069DB49CA1F6A0013.rev'
public and secret key created and signed.

pub   rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
       772ABE9DF4E0636C55B7C3F069DB49CA1F6A0013
uid    [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniversit
y.ac.id>
sub    rsa3072 2026-06-10 [E] [expires: 2028-06-09]

ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-06-09
/home/ian/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
       772ABE9DF4E0636C55B7C3F069DB49CA1F6A0013
uid    [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniversit
y.ac.id>
sub    rsa3072 2026-06-10 [E] [expires: 2028-06-09]

ian@ian-VirtualBox:~/Dinastian_2311104058$
```

b. Hasil kunci publik



The screenshot shows a terminal window with the following commands and output:

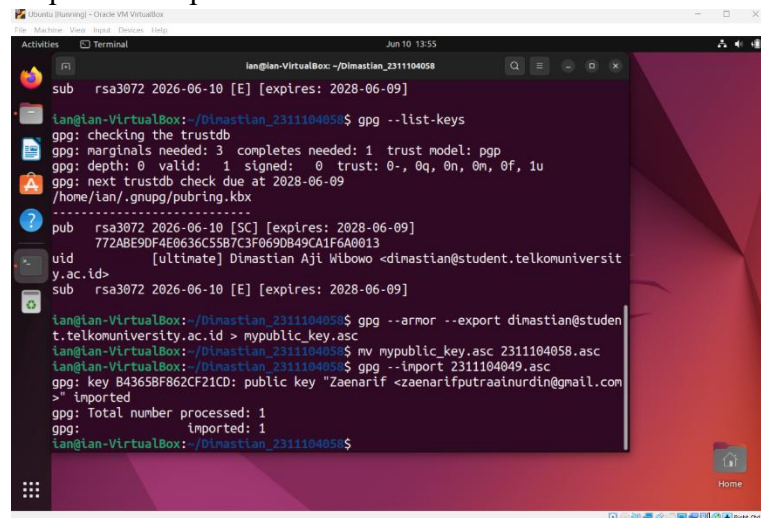
```
pub rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
772ABE9DF4E0636C55B7C3F069DB49CA1F6A0013
uid Dinastian Aji Wibowo <dinastian@student.telkomuniversit
y.ac.id>
sub rsa3072 2026-06-10 [E] [expires: 2028-06-09]

ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-06-09
/home/ian/.gnupg/pubring.kbx
-----
pub rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
772ABE9DF4E0636C55B7C3F069DB49CA1F6A0013
uid [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniversit
y.ac.id>
sub rsa3072 2026-06-10 [E] [expires: 2028-06-09]

ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --armor --export dinastian@studen
t.telkomuniversit
ian@ian-VirtualBox:~/Dinastian_2311104058$ mv mypublic_key.asc 2311104058.asc
ian@ian-VirtualBox:~/Dinastian_2311104058$
```

c. Export kunci publik

d. Import kunci publik



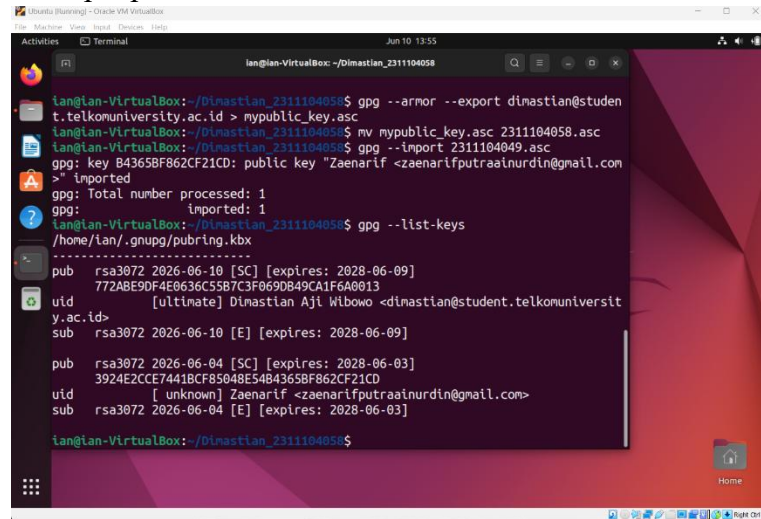
The screenshot shows a terminal window with the following commands and output:

```
sub rsa3072 2026-06-10 [E] [expires: 2028-06-09]

ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-06-09
/home/ian/.gnupg/pubring.kbx
-----
pub rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
772ABE9DF4E0636C55B7C3F069DB49CA1F6A0013
uid [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniversit
y.ac.id>
sub rsa3072 2026-06-10 [E] [expires: 2028-06-09]

ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --armor --export dinastian@studen
t.telkomuniversit
ian@ian-VirtualBox:~/Dinastian_2311104058$ mv mypublic_key.asc 2311104058.asc
ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --import 2311104049.asc
gpg: key B4365BF862CF21CD: public key "Zaenarif <zaenarifputraalnurdin@gmail.com
>" imported
gpg: Total number processed: 1
gpg: imported: 1
ian@ian-VirtualBox:~/Dinastian_2311104058$
```

e. Enkripsi pesan

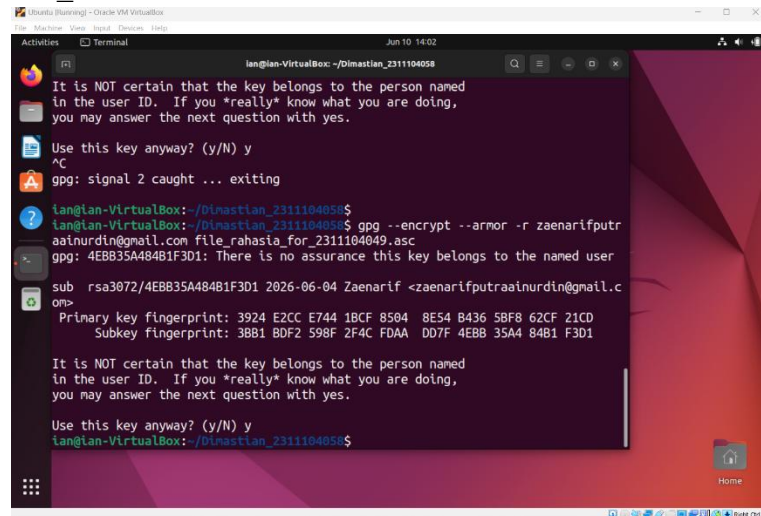


```
ian@ian-VirtualBox: ~/Dinastian_2311104058
ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --armor --export dinastian@studen
t.telkomuniversity.ac.id > mypublic_key.asc
ian@ian-VirtualBox:~/Dinastian_2311104058$ mv mypublic_key.asc 2311104058.asc
ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --import 2311104049.asc
gpg: key B4365BF862CF21CD: public key "Zaenarif <zaenarifputraainurdin@gmail.com
>" imported
gpg: Total number processed: 1
gpg:      imported: 1
ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --list-keys
/home/ian/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
      772ABE9DF4E0636C55B7C3F069DB49CA1F6A0013
uid   [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomunivsit
y.ac.id>
sub   rsa3072 2026-06-10 [E] [expires: 2028-06-09]

pub   rsa3072 2026-06-04 [SC] [expires: 2028-06-03]
      3924E2CCE7441BCF85048E54B4365BF862CF21CD
uid   [ unknown] Zaenarif <zaenarifputraainurdin@gmail.com>
sub   rsa3072 2026-06-04 [E] [expires: 2028-06-03]

ian@ian-VirtualBox:~/Dinastian_2311104058$
```

f. file_rahasia



```
ian@ian-VirtualBox: ~/Dinastian_2311104058
It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
^C
gpg: signal 2 caught ... exiting

ian@ian-VirtualBox:~/Dinastian_2311104058$
ian@ian-VirtualBox:~/Dinastian_2311104058$ gpg --encrypt --armor -r zaenarifputr
aainurdin@gmail.com file_rahasia_for_2311104049.asc
gpg: 4EBB35A484B1F3D1: There is no assurance this key belongs to the named user

sub   rsa3072/4EBB35A484B1F3D1 2026-06-04 Zaenarif <zaenarifputraainurdin@gmail.c
om>
Primary key fingerprint: 3924 E2CC E744 1BCF 8504 8E54 B436 5BF8 62CF 21CD
Subkey fingerprint: 3BB1 BDF2 598F 2F4C FDAA DD7F 4EBB 35A4 84B1 F3D1

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
ian@ian-VirtualBox:~/Dinastian_2311104058$
```